

ANANTAPUR, India

WMO: 432370

Lat: 14.583N

Lon: 77.633E

Elev: 1194

StdP: 14.07

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	61.7	64.0	39.9	37.8	89.6	43.5	43.5	88.4	11.9	78.6	10.8	79.1	0.5	90	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	21.5	104.9	74.3	103.3	74.1	101.6	73.5	80.9	94.4	79.4	92.4	78.3	90.9	4.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.6	150.0	87.7	76.3	143.5	85.4	75.3	138.6	83.8	45.7	94.8	44.1	92.7	42.8	91.0	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	18.1	15.6	DB	56.4	107.6	2.6	1.7	54.5	108.9	53.0	109.9	51.6	110.8	49.7	112.1	(4)
(5)			WB	52.0	84.2	2.3	2.0	50.4	85.6	49.0	86.8	47.8	87.9	46.1	89.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.1	76.5	81.0	87.2	91.3	90.8	86.5	84.2	83.2	82.5	81.3	77.9	75.4	(6)	
	DBStd	5.69	2.81	3.14	3.33	2.90	3.37	2.84	2.20	2.18	2.28	2.56	2.70	2.57	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	12097	820	868	1153	1239	1265	1095	1061	1029	975	969	837	786	(10)	
	CDD65	6622	355	448	688	789	800	645	596	564	525	504	387	321	(11)	
	CDH74	82746	3405	5528	10227	12518	12205	8552	7034	6338	5765	5180	3378	2616	(12)	
(13)	CDH80	43076	1302	2855	6291	8285	7863	4485	3098	2614	2353	2046	1112	772	(13)	
(14)	Wind	WSAvg	5.3	3.3	3.5	3.4	3.7	6.1	9.0	10.5	9.4	5.5	3.0	3.1	(14)	
(15) Precipitation	PrecAvg	23.2	0.2	0.1	0.3	1.0	2.7	2.6	2.3	4.1	4.7	3.8	1.1	0.3	(15)	
	PrecMax	42.2	2.4	1.1	2.5	3.1	8.3	8.5	8.2	13.1	12.0	8.4	4.1	2.2	(16)	
	PrecMin	11.6	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.4	0.2	0.0	0.0	0.0	(17)	
	PrecStd	6.4	0.4	0.2	0.6	0.9	2.0	2.1	2.2	3.0	3.1	2.2	1.3	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.5	99.5	104.9	107.5	106.8	103.1	97.0	96.0	95.4	94.9	90.9	90.6	(19)	
		MCWB	67.4	69.2	72.5	75.9	74.8	75.5	74.6	74.9	74.5	72.2	71.2	69.9	(20)	
	2%	DB	90.3	96.7	102.5	105.1	104.8	99.1	94.8	93.6	93.4	92.4	89.0	87.8	(21)	
		MCWB	67.3	68.7	71.9	73.9	75.2	76.4	74.6	74.4	74.1	72.7	71.1	68.5	(22)	
	5%	DB	88.5	94.6	100.9	103.6	103.2	96.9	93.1	91.9	91.7	90.6	87.5	85.9	(23)	
		MCWB	67.1	68.0	70.7	73.6	75.3	75.9	74.6	74.1	74.1	73.1	70.6	67.9	(24)	
	10%	DB	86.7	92.5	99.0	101.8	101.2	94.7	91.3	90.2	89.9	88.9	86.0	84.2	(25)	
		MCWB	66.6	67.2	70.0	73.3	75.2	75.7	74.4	74.0	74.3	73.4	70.5	67.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.8	76.3	80.1	82.1	83.6	82.1	78.6	78.9	79.9	79.1	76.9	74.6	(27)	
		MCDB	84.5	88.6	95.6	99.1	99.3	92.9	89.1	88.3	88.8	87.4	85.4	83.3	(28)	
	2%	WB	72.1	73.9	77.4	79.2	81.4	79.9	77.2	77.4	78.4	77.6	75.4	73.3	(29)	
		MCDB	82.8	86.4	92.0	96.4	96.2	91.1	88.6	86.5	87.0	86.0	82.9	81.3	(30)	
	5%	WB	70.9	72.5	75.8	77.7	79.6	78.4	76.3	76.6	77.4	76.8	74.5	72.1	(31)	
		MCDB	81.5	85.3	90.8	95.0	94.2	90.3	87.7	85.6	85.7	85.0	81.7	79.8	(32)	
	10%	WB	69.8	71.1	74.4	76.5	78.2	77.4	75.5	75.8	76.5	76.0	73.7	71.0	(33)	
		MCDB	80.4	84.6	90.2	93.8	92.8	89.4	86.7	84.9	84.7	83.8	80.8	78.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	23.3	23.7	21.5	20.3	16.4	14.5	14.0	14.4	15.0	16.1	19.0	(35)	
		MCDBR	23.7	26.2	25.4	23.1	22.3	19.3	16.9	16.4	17.4	18.4	18.5	21.7	(36)	
	5% WB	MCWBR	7.4	8.7	9.4	8.8	7.4	4.8	4.0	3.8	4.8	5.5	5.9	7.1	(37)	
		MCWBR	18.8	20.7	22.2	21.4	20.5	17.0	15.0	14.0	14.4	14.3	13.5	15.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.435	0.446	0.516	0.561	0.586	0.518	0.513	0.497	0.487	0.527	0.474	0.464	(40)	
		taud	2.155	2.132	1.953	1.879	1.842	2.061	2.078	2.135	2.164	2.022	2.130	2.098	(41)	
	Ebn at Noon		271	274	258	245	235	249	251	258	261	248	257	258	(42)	
		Edh at Noon	47	50	61	66	67	53	52	50	49	55	48	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1673	1925	2074	2138	2038	1695	1483	1518	1608	1582	1501	1503	(44)	
	RadStd		63	72	65	56	77	115	97	67	113	137	109	81	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.21	+0.97	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.50	+0.58	N/A	+0.97	+0.41	N/A	N/A	N/A	+308	+310

Nomenclature: See separate page